

Apart from energy efficiency, what were the other prevalent themes at the summit?

Innovation is coming through very strongly. It is becoming very evident that the proposed \$400-billion market will not be achieved through me-too products, but very innovative products that address the problems in India and for India.

How has the electronics industry in India grown over the years?

In Pradeep Dhoot's session at the summit, he said when Videocon was started in the early 1980s, the industry was worth about \$100 million. Today, the industry is \$70 billion in size and it's talking about growing at four times this pace to reach about \$400 billion in the next eight years.

We are talking about growing from 160,000 engineers to about three million engineers. Whether it is in monetary terms or employment terms, the industry is growing by leaps and bounds. Underpinning that are two mega trends in India: 60 per cent of the workforce is below the age of 25, and the fundamentals of the economy are still fine. We have got a \$2-trillion economy growing at 8 per cent, which is very creditable in today's times.

What are the growing verticals?

Telecom is by far the biggest, so are consumer and office automation. We are also seeing increasing traction in industrial, automotive and medical electronics. You will also see new verticals opening up—like energy efficiency, which is becoming critical to the success of most companies.

Energy efficiency is a national imperative as we simply do not generate enough electricity. We are working towards products using much lesser energy to do the same tasks. In large data centres or mass-volume products like fans, a simple change in the design can considerably reduce the amount of energy consumed.

Could you explain how this energy saving can be achieved?

Last year, India made 500 billion fans. On an average, a fan consumes about 70 watts of power. A very simple variable digital logic controller circuitry built-in, which is about \$1 semiconductor component, can cut the fan's power consumption to 30 watts. So we can save more than half the amount of power consumed by 500 billion fans.

Also, there are LED lamps, 5-star rated air-conditioners, mobile phone circuits and mobile phone chargers that switch off automatically.

As a nation, we have 800 million cellphones. Ninety per cent of users do not switch off their mobile phone chargers these days. Standby current on the charger is 4 watts and at any point of time, 600 million chargers are connected. With very simple electronics, this can be reduced to half a watt.

Ninety-nine per cent of the people do not switch off the television from the wall socket, but with a remote. You still consume 40 watts of power, which can be cut down to 10 watts.

Is there space for more players in the Indian electronics market?

Absolutely, we have expanded and built on the government's projections. The government projection under the National Electronics Policy talks about 200 successful start-ups in that space. Going by the law of averages, we are talking about 2000 start-ups in that space and 200 of them surviving through mergers and acquisitions.

ISA has expanded that and expects to see 50 fabless semiconductor companies, each with a turnover of \$200 million. That is, there would be 500 start-ups, of which about 50 would consolidate.

Does India stand a chance of competing with China in terms of manufacturing?

I sure hope so. I do not intend to hand over three times the domestic market for semiconductors and electronics to the Chinese. Otherwise, we would not be doing what we are doing today. The issue is going to be about scale.

We have to give domestic companies as much impetus as possible. China is already at a high level and anything that they build further goes on that already very large base. So their per-unit transaction costs come down.

In India, that is not the case. Many parameters still have to be put into place. For example, if I want to build a fab, I have to invest in a 100 per cent power backup system, water treatment plants, transport infrastructure to bring my employees in and out of the plant, cafeteria facilities—all of which adds to the cost. In China, I can depend on the utility company to give me power and bus services, and there would be a series of hotels without having to invest into it.

Quoting from a completely different industry, I remember K. Ganesh of Tutor Vista saying that before he started his tutoring venture, he was not running a KPO, but a garage, taxi service, cafeteria, hotel service, admin service, etc—anything but running a KPO. The disability costs of doing a business in India are 19-22 per cent. These are the costs you will not incur elsewhere.

Finally, what's the current focus of ISA?

ISA is the voice of the electronics industry. We started seven years ago representing the semiconductor, embedded software space and allied areas like VLSI design, electronic design automation, embedded software, hardware board design and PCB. Sometime around 2009, we realised that just representing the heart of the electronics industry was not enough and that we have to represent the larger constituency of electronics, i.e., the electronics systems industry. So we started focusing on the electronic system design and manufacturing space.

ISA has been working hard to put out the message that India has to boost domestic manufacturing in electronics so that it doesn't face balance of payments issue in the near future. ●